SOFTWARE VALIDATION REPORT

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Date:	09/08/2024	09/08/2024	09/08/2024

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1. Introduction

This report presents the results of the validation of the VRCardio VHET360 software, hereinafter VRCardio , developed by Spika Tech, hereinafter SPIKA, specifically designed for the visualization and analysis of cardiac signals. The validation verifies that VRCardio meets the necessary functionality, reliability, and safety standards.

2. Scope

Validation covers all aspects of the VRCardio environment, focusing on three key areas: functionality, reliability, and security. It includes verification of all environment functions, detection of critical errors, and security assessment against known vulnerabilities.

3. Definitions

Validation : The process of evaluating whether a system or component meets requirements and functions as intended in its operating environment.

Installation Qualification (IQ): Verification that the software and all its components have been installed correctly in the target environment.

Operational Qualification (OQ) : Confirmation that the software operates correctly and that all major functionalities and modules function as expected.

Performance Qualification (PQ) : Ensuring that the software meets specific business requirements and operates according to user-defined performance and functionality criteria.

4. Validation plan

4.1. Goals

The main objectives of the VRCardio software validation are:

- Ensure that the software meets established functionality and performance requirements.
- Verify the reliability and stability of the software under different operating conditions.
- Evaluate software security against known and potential vulnerabilities.

4.2. Validation methods

Methods used for validation include:

- Installation and configuration testing.
- Basic and advanced functionality testing.
- Integration and communication tests with other systems.
- Performance tests under different loads.
- Security and attack resistance testing.

4.3. Acceptance criteria

The software will be considered accepted if it meets the following criteria:

- Correct installation and configuration without errors.
- Correct operation of all main functionalities.
- Successful integration with external systems and hardware.
- Adequate performance under different load conditions.
- Compliance with established safety standards.

4.4. Validated software version

With the completion of this software validation report, the following version of VRCardio is being carried out: 1.0.0.

5. Validation procedures

5.1. Installation Qualification (IQ)

5.1.1. Review of installation documentation

All the information available for downloading the software and getting started with creating a project has been compiled:

- Download link: There is no download link yet, until the app is publicly released.

- Installation: Unzip the .zip file and open the VRCardio-VHET360-Visualization.exe file.
- Minimum requirements:
    - Operating system: Windows 10 64-bit
    - Processor: 8-Core Intel Core i5 or equivalent
    - RAM: 8 GB or more
    - For non-VR use: NVIDIA GeForce GTX 1050 ti or equivalent
    - For use with VR: NVIDIA GeForce RTX 2070 or equivalent
    - Disk space: Minimum 2 GB of free space

After this, the software was downloaded and installed successfully.

5.1.2. Hardware and software testing

For the VRCardio version (1.0.0) the minimum requirements indicated for its correct operation are:

- Windows 10 or higher
- x64 architecture
- DX10, DX11 or DX12

The computers on which this software has been installed and tested have the following characteristics:

- Windows 10 or Windows 11
- x64 architecture
- DX12

5.1.3. Installation registration

Process	Staff	Incidents

Access to internal download Downloading and installing the software Login with a private development account Access to the developed software	David Pozo	None
Access to internal download Downloading and installing the software Login with a private development account Access to the developed software	Daniel Cámara	None
Access to internal download Downloading and installing the software Login with a private development account Access to the developed software	Javier Vico	None

5.1.4. File integrity validation

After each software download and installation, the local path folder was accessed to verify that all files existed and had a modification date consistent with the installation date, confirming that no files had been modified in any way after installation.

5.2. Operation Qualification (OQ)

5.2.1. Basic functionality tests

Test performed	Result	Confirmation
Open the application.	The application opens without errors.	The application has started loading and then opening.
Login.	The user logs in without any problem.	The user successfully accesses his account.

Log out.	The user logs out of his account.	The user successfully logs out and returns to the login screen.
Access an offline room.	The user creates an offline room and successfully logs in.	The user can interact with the room panels without any problem.
Upload a case	The user accesses the list of cases, loads one of them and it appears on the screen	The user can view and interact with the case without any problem.
View and manage cardiac signals.	The user accesses the signals and can view them without any problem.	The signals have been correctly represented in the graph.
Managing time	The user has been able to manage the session time using the controllers.	The time has responded correctly to user actions.
Use VR glasses	The user has successfully entered a room using VR glasses	The user views the scene correctly using VR glasses
Use holographic display	The user has successfully entered a room using a holographic display	The user visualizes the scene correctly using a holographic display
Access an online room	The user accesses an online room using the code	The user can view the online scene next to the other users' pointers.
Modify panel position	The user has been able to move and rotate the room panels without any problem.	The panels have been modified in position and rotation.
Hide and show panels	The user has hidden several	Panels have appeared and disappeared at the user's

	panels without any problem.	whim.

5.2.2. Integration tests

Test performed	Result	Confirmation
Database integration	Data is stored and retrieved correctly.	Results of queries and updates: • Test query performed in 10ms, accurate data retrieval. • Registry update completed in 20ms, successful update confirmation.
Communication with remote servers	Communication is stable and error-free.	Communications records: • HTTP request sent • Response received in 200 ms, no connection errors.
Data synchronization between devices	Data is synchronized properly.	Confirmation of data synchronized correctly between device A and B.
User authentication	Users are authenticated successfully.	Authentication completed in 50ms, no errors.
Interaction with an API	The API responds correctly.	Response received in 300 ms with correct patient data
Use of cloud services	Cloud services run smoothly.	File uploaded and accessible in 5 seconds,

		without errors.
Interaction with external hardware	The hardware responds appropriately.	Data readings received in 100 ms, with no communication errors.
Real-time data synchronization	Synchronization is fast and error-free.	Confirmation of successful real-time synchronization.
User session management	Sessions are managed correctly.	Login and logout records: • User login at 12:00 PM. • Session closed at 12:30 PM. • Session managed without problems.
Integration with storage services	Data is stored and retrieved seamlessly.	Data stored in the database. Data recovery completed in 2 seconds, no errors.
Application monitoring	Monitoring is performed without interruptions.	Confirmation of monitoring data received continuously without interruptions.

5.2.3. Performance tests

Test performed	Result	Confirmation
Maximum load of simultaneous users	The application supports maximum load without degradation.	Response time and resource utilization records: • Number of simultaneous users: 50

		• Average response time: 200 ms • CPU Utilization: 70% • usage : 65%
Stress test	The system recovers properly after stress.	Stress test results: • Application subjected to heavy file load • Full recovery to normal state in seconds.
Latency test	Latency remains within acceptable limits.	Latency test results: • Average latency: 150 ms • Maximum latency: 300 ms • Tests performed under different network conditions.
Concurrency test	The system handles multiple concurrent requests.	Concurrency test results: • 1,000 concurrent applications. • All applications processed successfully without errors.
Memory Usage Test	The application handles memory usage properly .	Memory usage during load test: 65% No memory leaks detected.

CPU Usage Tests	The application handles CPU usage properly.	CPU usage during load test: 75%
Bandwidth testing	The application works well with different bandwidths.	Bandwidth used: 10 Mbps, 50 Mbps, 100 Mbps. Stable performance
Response testing on different networks	The application responds appropriately on different networks.	Test results on various networks: • Tested networks: 3G, 4G, 5G, Wi-Fi. • Average response time on 3G: 500 ms • Average response time on 4G: 300 ms • Average response time on 5G and Wi-Fi: 100 ms
Load time tests	Loading times are acceptable.	Initial load time: 6 seconds
Script runtime testing	Scripts run within the expected time.	Script execution time logs: • Average execution time: 50 ms • execution time : 200 ms
Background resource usage tests	The app handles background resources well.	Resource usage results: • Background CPU usage: 30% • Background memory

		usage: 35%

5.2.4. Security check

Test performed	Result	Confirmation
Proof of unauthorized access	The system blocks unauthorized access.	Logs of access attempts and blockages: • All attempts were successfully blocked
SQL Injection Testing	The system is immune to SQL injections.	No SQL injection was successful.
Cross-Site Scripting (XSS) Testing	No XSS vulnerabilities found.	No XSS vulnerabilities were found.
Access control tests	Access controls are working properly.	Users with different roles tested. All accesses were properly controlled according to the permits.
Data integrity testing	Data integrity is guaranteed.	All data remained intact during the test.

5.3. Performance Qualification (PQ)

5.3.1. User acceptance testing

Test performed	Result	Confirmation
Installing and running the software	Users download, install and run the application without any problems.	Feedback on the ease of configuration and functionality of the

		parameters: • User A: "The process was very simple." • User B: "I was able to install the app without any problems." • User C: "Downloading and installing the app is easy."
Using display panels	Users use data visualization dashboards to analyze patterns.	Comments on the effectiveness and ease of use of the display panels: • User A: "The panels are useful and easy to use." • User B: "It's very convenient to use the panels to change the display." • User C: "I enjoyed the experience of using the panels ."
Users evaluate the navigation and usability of the software	Users evaluate the navigation and usability of VRCardio software.	Feedback on the user interface and ease of use: • User A: "It has been intuitive and comfortable." • User B: "The application has good usability."

		• User C: "Easy to use and easy to handle."

5.3.2. Critical scenario testing

Test performed	Result	Confirmation
Data recovery after a failure	A system failure is simulated and data recovery is verified.	Recovery time and accuracy of recovered data: • Failure simulation performed. • Data recovery completed. • Recovery time: 1 minute. • All data was recovered accurately and without loss.
Real-time analysis	Real-time data analysis is evaluated during critical operations	Performance and accuracy of real-time analysis: • Recorded performance: Average response time of 50 ms. • Analysis accuracy: 99.5%
High demand scenarios	High-demand scenarios are simulated and software performance is evaluated.	Software performance under high demand: • Test performed under high

		processing load • The software handled the load without significant performance drops. • Average response time: 200 ms.
Managing large volumes of data	The software's ability to manage and analyze large volumes of data is evaluated.	Performance and accuracy in analyzing large volumes of data: • Test performed with 10 GB of data. • Recorded performance: Processing completed in 30 minutes. • Analysis accuracy: 98.7%

5.3.3. Evaluating user documentation

Test performed	Result	Confirmation
User manual review	Verification that the user manual covers all software features.	All major features are covered.
Setup Guide	The setup guide is ensured to be clear and complete.	The guide covers all the steps necessary for initial setup.

Good Practice Guide	The good practices guide is verified to be complete and understandable.	The guide is comprehensive and covers all recommended best practices.
Video tutorials	Video tutorials are said to be clear and educational.	The tutorials are clear and well structured.

6. Results and conclusions

After conducting extensive installation (IQ), operational (OQ), and functional (PQ) testing, it was concluded that VRCardio meets the standards required for medical use. The results obtained at each stage of validation indicate the following:

- Successful Installation: VRCardio installation proceeded without incident, meeting the established hardware and software requirements. All files were installed correctly and their integrity was verified.
- Successful Operation: Basic functionality, integration, performance, and security tests show that VRCardio operates correctly under various conditions. No significant errors were found, and all core features responded as expected.
- Proper Operation: User acceptance testing and critical scenarios validate that the environment is suitable for developing cardiac signal visualization and analysis applications. Users found the software intuitive and effective for their purposes.

Based on the data collected, VRCardio is approved for use, ensuring that it meets the necessary functionality, reliability, and safety criteria.

7. Version history

VERSION	REASON FOR CHANGE	DATE
01	Initial edition	July 2024

8. Conformity of the person responsible for validation

State:

☐According

☐Not compliant

Name:

Date:

Signature: